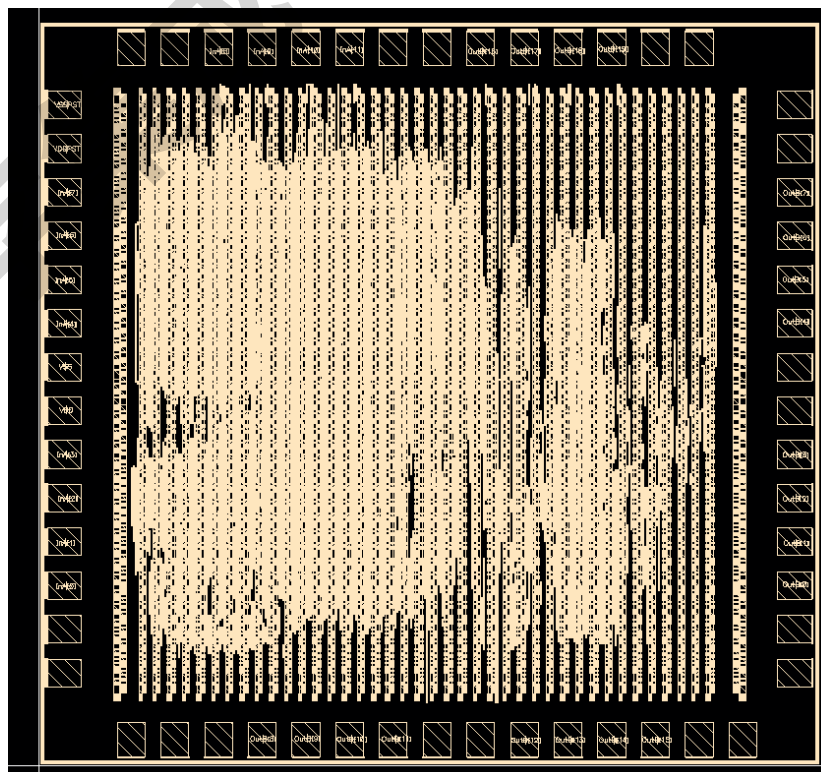


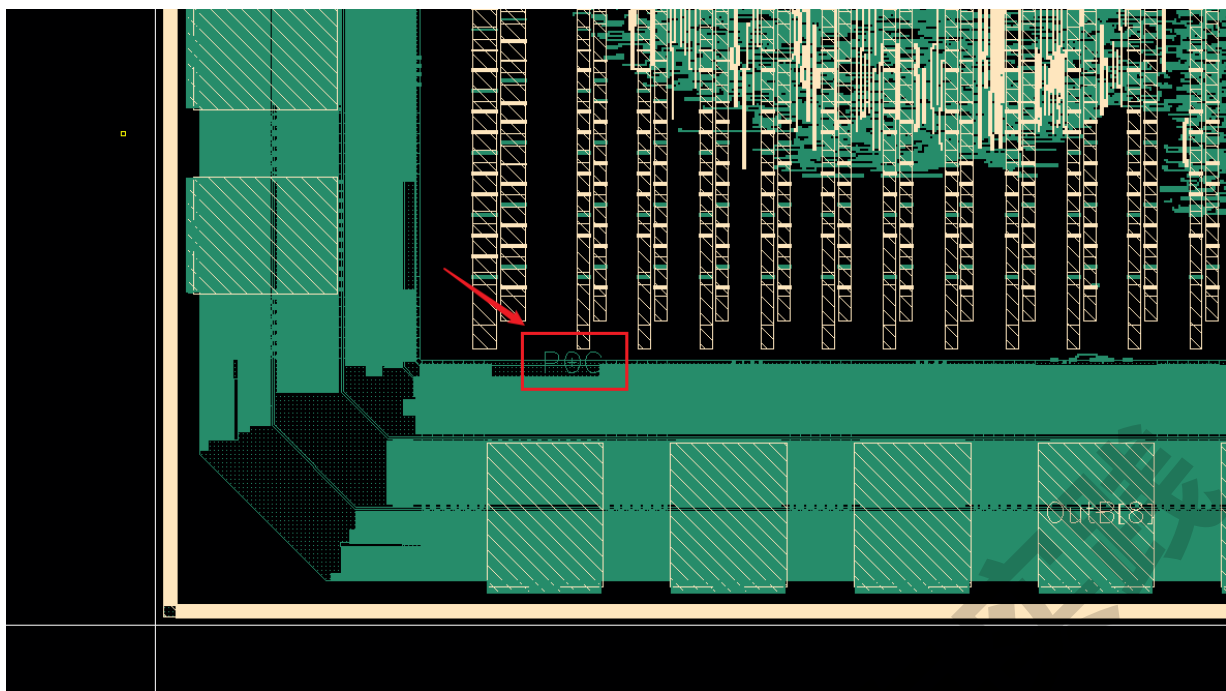
LVS 常见 ERROR 与解决方案

1. Missing port

LAYOUT NAME	SOURCE NAME
✖ Discrepancy #5 in RV64_SealRing_PAD_Dum	
** missing port **	VDDPST on net: VDDPST
✖ Discrepancy #6 in RV64_SealRing_PAD_Dum	
** missing port **	VDD on net: VDD
✖ Discrepancy #7 in RV64_SealRing_PAD_Dum	
** missing port **	VSSPST on net: VSSPST
✖ Discrepancy #8 in RV64_SealRing_PAD_Dum	
** missing port **	VSS on net: VSS
✖ Discrepancy #9 in RV64_SealRing_PAD_Dum	
** missing port **	POC on net: POC
✖ Discrepancy #10 in RV64_SealRing_PAD_Dum	
** missing port **	INA[11] on net: INA[11]
✖ Discrepancy #11 in RV64_SealRing_PAD_Dum	
** missing port **	INA[10] on net: INA[10]
✖ Discrepancy #12 in RV64_SealRing_PAD_Dum	
** missing port **	INA[9] on net: INA[9]
✖ Discrepancy #13 in RV64_SealRing_PAD_Dum	
** missing port **	INA[8] on net: INA[8]

在芯片 IO 上添加 Label, 打在 Metal6 pin 上。注意要加在最顶层的 layout 上才会起作用。另外 POC 较为特殊, 需要将 Label 打在 Metal3 pin 上, 并且打在 IO 内部环上。





数字集成电路设计

2. Property Error

Cell RV64_SealRing_PAD_Dum (1000 Property Errors)			
LAYOUT NAME	SOURCE NAME		
✖ Discrepancy #1042 in RV64_SealRing_PAD_Dum			
X34/X4746/M11(344,115,510,620) MP(P)	Xrvcpu/Xcore/cpu_core/XFE_RC_2620_0/M2 MP(P)		
l: 0.183782 u	l: 0.18 u	2.1%	
w: 3.60426 u	w: 3.6 u	0.118%	
✖ Discrepancy #1043 in RV64_SealRing_PAD_Dum			
X34/X4746/M13(342,695,510,620) MP(P)	Xrvcpu/Xcore/cpu_core/XFE_RC_2620_0/M3 MP(P)		
l: 0.183782 u	l: 0.18 u	2.1%	
w: 3.60426 u	w: 3.6 u	0.118%	
✖ Discrepancy #1044 in RV64_SealRing_PAD_Dum			
X34/X4746/M4(342,390,512,100) MN(N)	Xrvcpu/Xcore/cpu_core/XFE_RC_2620_0/M1 MN(N)		
l: 0.18567 u	l: 0.19 u	2.28%	
w: 2.40426 u	w: 2.4 u	0.177%	
✖ Discrepancy #1045 in RV64_SealRing_PAD_Dum			
X34/X4746/M2(343,980,512,110) MN(N)	Xrvcpu/Xcore/cpu_core/XFE_RC_2620_0/M0 MN(N)		
l: 0.185718 u	l: 0.19 u	2.25%	
w: 2.38426 u	w: 2.38 u	0.179%	
✖ Discrepancy #1046 in RV64_SealRing_PAD_Dum			
X34/X4746/M15(339,990,509,985) MP(P)	Xrvcpu/Xcore/cpu_core/XFE_RC_2620_0/M9 MP(P)		
l: 0.18471 u	l: 0.18 u	2.62%	
w: 1.44713 u	w: 1.45 u	0.198%	
✖ Discrepancy #1047 in RV64_SealRing_PAD_Dum			
X34/X4746/M6(339,990,512,440) MN(N)	Xrvcpu/Xcore/cpu_core/XFE_RC_2620_0/M8 MN(N)		
l: 0.187012 u	l: 0.19 u	1.57%	
w: 0.972132 u	w: 0.97 u	0.22%	
✖ Discrepancy #1048 in RV64_SealRing_PAD_Dum			

这是由于 layout 抽取的 MOS 管栅长与宽与理想值有一定的偏差，LVS command file 默认的误差容忍范围为 0%，修改 LVS command file 将误差容忍范围调整为 5%。

```
1946 #IFDEF RC_DECK
1947 TRACE PROPERTY MN(N) L L 5
1948 TRACE PROPERTY MN(N) W W 5
1949 TRACE PROPERTY MN(N1) L L 5
1950 TRACE PROPERTY MN(N1) W W 5
1951 TRACE PROPERTY MN(N2) L L 5
1952 TRACE PROPERTY MN(N2) W W 5
1953 TRACE PROPERTY MN(NA) L L 5
1954 TRACE PROPERTY MN(NA) W W 5
1955 TRACE PROPERTY MN(NB) L L 5
1956 TRACE PROPERTY MN(NB) W W 5
1957 TRACE PROPERTY MN(ND) L L 5
1958 TRACE PROPERTY MN(ND) W W 5
1959 TRACE PROPERTY MN(NL) L L 5
1960 TRACE PROPERTY MN(NL) W W 5
1961 TRACE PROPERTY MN(NN) L L 5
1962 TRACE PROPERTY MN(NN) W W 5
1963 TRACE PROPERTY MP(P) L L 5
1964 TRACE PROPERTY MP(P) W W 5
1965 TRACE PROPERTY MP(PA) L L 5
1966 TRACE PROPERTY MP(PA) W W 5
1967 TRACE PROPERTY MP(PD) L L 5
1968 TRACE PROPERTY MP(PD) W W 5
1969 #ENDIF
```

3. CASE SENSITIVE

[illegible]

如上图所示，发现有部分连接关系错误，有可能是由于大小写的问题。导出的 SPI 文件默认是大小写敏感的，例如上图中的例子中 N405 和 n405 本质上是两根信号线，但是 LVS 中大小写不敏感，导致将 N405 和 n405 认为为一个信号线，导致出现连接错误。因此需要在 LVS command file 中加入下面三句语句

SOURCE CASE YES

LAYOUT CASE YES

LVS COMPARE CASE YES

4. IO 内部的 LVS 错误(可 waive)

在 PDDW0204CDG 内部出现的下述 ERROR 可以 waive

